JAVA SCRIPT

1️⃣ Basics

- Introduction to JavaScript
- How JavaScript works
- Variables (var, let, const)
- Scope (Global, Block, Function)
- Data Types (Primitive & Non-Primitive)
- Type Conversion & Coercion
- Operators (Arithmetic, Logical, Comparison, Ternary)

2️⃣ Control Statements

- if
- else
- else if
- switch
- for
- while
- do…while
- for…of
- for…in

3️⃣ Functions

- Normal Function
- Function Expression
- Arrow Function
- Anonymous Function
- IIFE
- Parameters & Arguments
- Default Parameters
- Return Statement

- Callback Functions

4️⃣ Arrays

- Creating Arrays
- Array Indexing
- Array Length
- Array Methods:
  - push, pop
  - shift, unshift
  - map
  - filter
  - reduce
  - forEach
  - find
  - includes
  - sort

5️⃣ Objects

- Object Creation
- Accessing Properties
- Object Methods
- this keyword
- Destructuring
- Spread Operator (...)
- Object Methods:
  - Object.keys()
  - Object.values()
  - Object.entries()

6️⃣ Strings & Numbers

- String Methods:
  - length
  - toUpperCase()
  - toLowerCase()
  - slice()
  - substring()
  - replace()
  - split()
- Number Methods:
  - parseInt()
  - parseFloat()
- Math Object:
  - Math.random()
  - Math.floor()
  - Math.ceil()

7️⃣ DOM (Document Object Model)

- DOM Tree
- getElementById()
- getElementsByClassName()
- querySelector()
- querySelectorAll()
- Changing HTML content
- Changing CSS using JS

8️⃣ Events

- click
- change
- submit
- mouseover

- mouseout
- keydown
- keyup
- Event Listener
- Event Bubbling & Capturing

9️⃣ BOM (Browser Object Model)

- window
- alert()
- prompt()
- confirm()
- setTimeout()
- setInterval()
- location
- history

🔟 ES6 & Modern JavaScript

- let & const
- Arrow Functions
- Template Literals
- Destructuring
- Spread & Rest Operator
- Default Parameters
- Optional Chaining
- Modules (import, export)

1️⃣1️⃣ Asynchronous JavaScript

- Synchronous vs Asynchronous
- Callbacks
- Promises
- async / await

- Error Handling (try…catch)

1️⃣2️⃣ API Handling

- fetch()
- JSON
- HTTP Methods (GET, POST, PUT, DELETE)

1️⃣3️⃣ Advanced JavaScript

- Hoisting
- Closures
- Scope Chain
- Execution Context
- Prototype
- Prototypal Inheritance
- call(), apply(), bind()
- Shallow Copy vs Deep Copy

1️⃣4️⃣ JavaScript for React (Must Know)

- ES6 usage in React
- Array methods in JSX
- Object destructuring
- Event handling
- Async JS with React

Practice website:

- **W3Schools**
- **JavaScript.info**
- **HackerRank**
- **Codewars**
- **LeetCode**
- **FreeCodeCamp**
- **Frontend Mentor**
- **JSFiddle**
- **CodePen**

Must practice types

- Arrays (sum, max, filter, map)
- String problems (reverse, count)
- Objects (loop, access, update)
- DOM (button click, form validation)
- Async (fake API fetch, promise)